



PTT PHENOL COMPANY
LIMITED

PROCESS SPECIFICATION

PREFLASH COLUMN STEAM HEATER (E-2303)

Docu. No. : 14780-8120-PS-E2303
Project No. : 120117
Rev. No. : Z1
Date : MAY 11, 2016
Page : 1 / 10

TITLE :

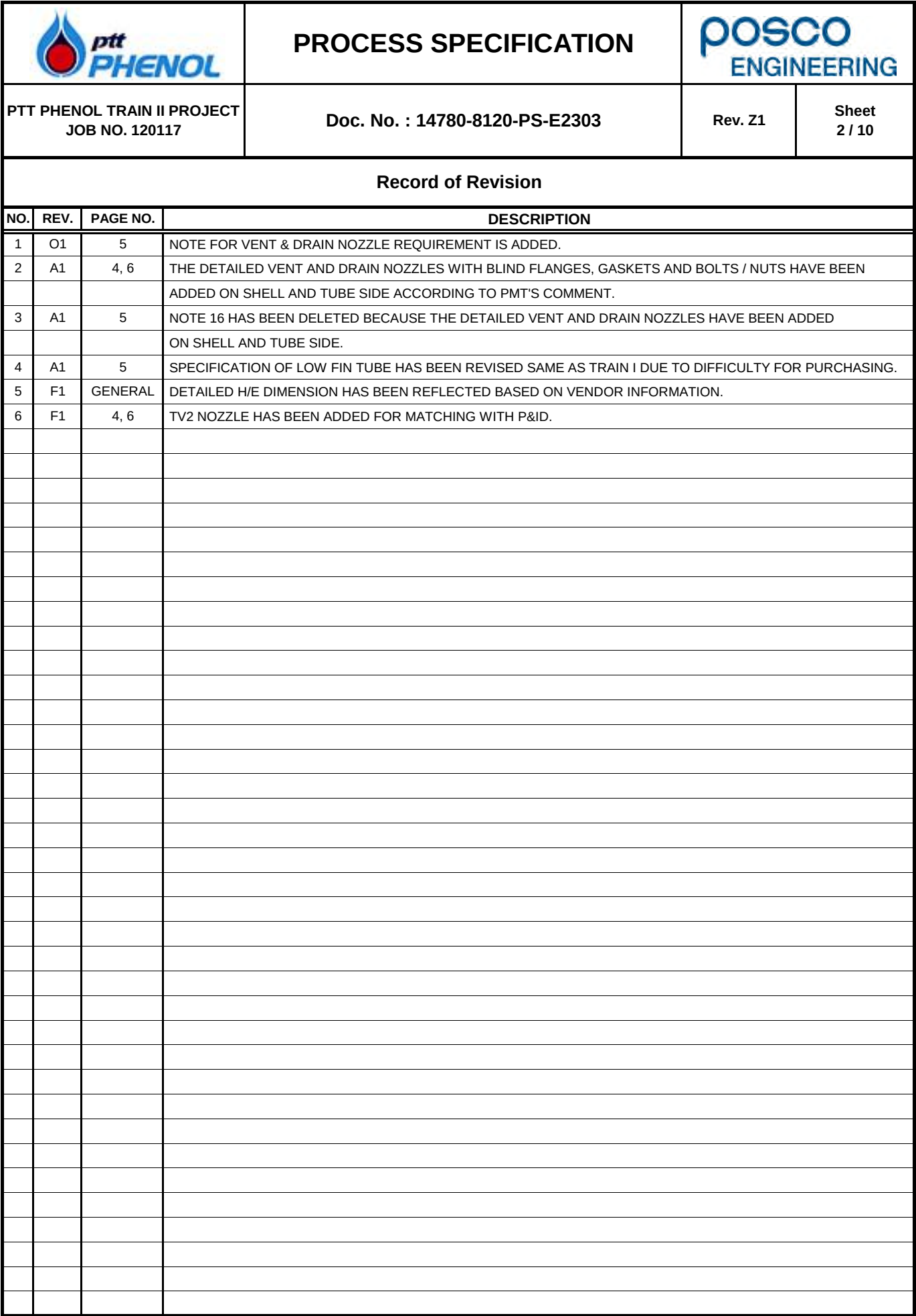
PREFLASH COLUMN STEAM HEATER

(E-2303)

DISCIPLINE : PROCESS
DOCUMENT NO. : 14780-8120-PS-E2303
PROJECT NAME : PTT Phenol Train II Project
PROJECT NO. : 120117
OWNER : PTT Phenol Company Limited (PPCL)

Z1	MAY 11, 2016	AS BUILT		S.W.Park	J.S.Chung	- S.W.Lee
F1	MAY 28, 2014	FINAL ISSUE		S.W.Park	J.S.Chung	- S.W.Lee
A1	OCT 8, 2013	APPROVED FOR DESIGN		S.W.Park	J.S.Chung	- S.W.Lee
O1	JUN 27, 2013	COMMENT ISSUE		S.W.Park	J.S.Chung	- S.W.Lee
REV. NO.	DATE	DESCRIPTION		PREP'N	REVIEW	QA REVIEW APPROVAL
REV. NO.	PREPARATION DEPT.	PREP'N	REVIEW	QA REVIEW		APPROVAL
Z1	PROCESS	S.W.Park	J.S.Chung	-		S.W.Lee
		MAY 11, 2016	MAY 11, 2016	-		MAY 11, 2016

This Document is the property of POSCO Engineering Co., Ltd. Therefore, it shall not be released to Any third party without permission of an authorized personnel of the POSCO Engineering Co., Ltd.





SHELL AND TUBE EXCHANGER THERMAL DATA SHEET

System of Units

Metric

Doc. No. : 14780-8120-PS-E2303

Rev. No. : Z1

Item No. : E-2303

Page : 3 / 10

Case		Design	TEMA Type (Note 3, 4)	C_U	Orientation	Horizontal	Rev
Number of Units		1	Shells per Unit	1	Shells in Series	1	
Surface per Unit	m ²	996.5	Surface per Shell	m ²	996.5	Shells in Parallel	1
		(Note 8)			(Note 8)		
PERFORMANCE OF ONE UNIT							
		In	SHELLSIDE		Out	In	TUBESIDE
				Out			Out
Fluid Name		Cumene & CHP		S1.5 Steam			
Total Flowrate (Note 2)		kg/h	149940 × 1.25		11388 × 1.25		
Vapour		kg/h	0	64718 × 1.25	11388 × 1.25	0	
Liquid		kg/h	149940 × 1.25	85222 × 1.25	0	11388 × 1.25	
Fluid Condensed		kg/h	0		11388 × 1.25		
Fluid Evaporated		kg/h	64718 × 1.25		0		
LIQUID PROPERTIES							
		SEE PAGES 9, 10					
Density	kg/m ³						
Viscosity	cP						
Specific Heat	kcal/kg.°C						
Thermal Conductivity	kcal/h.m.°C						
Surface Tension	dyne/cm						
Pour Point Temperature	°C						
VAPOUR PROPERTIES							
		SEE PAGES 9, 10					
Density	kg/m ³						
Viscosity	cP						
Specific Heat	kcal/kg.°C						
Thermal Conductivity	kcal/h.m.°C						
Hydrate Temperature	°C						
THERMAL AND HYDRAULIC DATA							
Operating Temperature	°C	60.2	70.1	120.5	114.5		
Operating Pressure	kg/cm ² g	-0.9210	(Note 5)	0.6646			
Number of Passes per Shell		1		2			
Average Velocity	m/s	10.0		21.7			
Pressure Drop (Allow / Calc)	kg/cm ²	0.0600	0.092 (Note 6)	Nil	0.118		
Fouling Factor	h.m ² .°C/kcal	0.0004		0.0001			
Duty (Note 2)	kcal/h × 10 ⁶	6.075 × 1.25		MTD	°C	34.3	
U-value Service/Clean	kcal/h.m ² .°C	223	268	Over Surface	%	0.4	
		(Note 7)					
DESIGN CONDITIONS							
		SHELLSIDE		TUBESIDE			
Design pressure max/min	kg/cm ² g	3.5		7.0			
Full Vacuum		Required	at 195 °C	Required	at 120 °C		
Test Pressure	kg/cm ² g	To Code		To Code			
Design Temperature max/min	°C	250	15	195	15		
Corrosion Allowance	mm	TBA		3			
Insulation Details		Heat Conservation		Heat Conservation			
Hydrogen Service		No		No			
Sour Service		No		No			
Lethal Service		No		No			
Cyclic Service		No		No			
Cleaning Method		Mechanical		Mechanical			
Stress Relief for Process Reasons		No		No			
Heat Release Curve		Yes		No			

SHELL AND TUBE EXCHANGER THERMAL DATA SHEET

System of Units

Metric

Doc. No. : 14780-8120-PS-E2303

Rev. No. : Z1

Item No. : E-2303

Page : 4 / 10

CONSTRUCTION OF ONE SHELL

SHELL	Shell I.D.	mm		Channel I.D. (If different)	mm	1160
TUBES	Number of Tubes		1464 (Note 13)	Tube Length	mm	4800
	Tube O.D.	mm	19.05	Tube Thickness	mm	1.651 (Note 11)
	Tube Gauge	BWG	16	Tube Wall Basis		Average
	Tube Pitch	mm	25.4	Tube Layout Angle	deg.	90
	Tube-Tubesheet Joint		Strength Welded (Note 12)	Tube-Baffle Diam. Clearance	mm	0.38
BAFFLES	Number of Baffles		5	Baffle Type		Support
	Baffle Cut	%		Baffle Spacing	mm	960
	Baffle Cut Orientation		Supports Only	Double Seg. Baffle Overlap	mm	
	Baffle Thick	mm	9.5	Baffle-Shell Diam. Clearance	mm	6.4
BUNDLE	Impingement Protection		None	Tube Supports		None
	Pairs of Sealing Strips		No	Sealing Strip Details		N.A.
	Passlane Sealing		No	Passlane Sealing Details		N.A.
	U-Bend Diam.	mm	57.15	U-Bend Orientation		Vertical
	U-Bend Area Included		No	U-Bend Support Details		(Note 14)
	Longitudinal baffle		No	Longitudinal Baffle Seal		N.A.

CONNECTIONS

SHELLSIDE						TUBESIDE					
Service	No.	N.B.		Rating	Facing	Service	No.	N.B.		Rating	Facing
						Inlet T1	1	18	Inch	150	RFWN
						Outlet T2	1	4	Inch	150	RFWN
						Vent TV1	1	3/4	Inch	150	RFLWN
						Drain TD	1	3/4	Inch	150	RFLWN
						Vent TV2	1	2	Inch	150	RFLWN

MATERIALS

Tubes	304L Stainless Steel
Shell / Shell Cover	N.A.
Floating Head Cover	N.A.
Channel / Channel Cover	304L Stainless Steel (UOP Std 3-15) / Killed Carbon Steel (UOP Std 3-11)
Tubesheet, Stationary	304L Stainless Steel
Tubesheet, Floating	N.A.
Baffles / Tube Supports	304 Stainless Steel
Nozzles / Nozzle Flange	Shellside: N.A.
Nozzles / Nozzle Flange	Tubeside: 304L Stainless Steel
Bolt and nuts	Per Project Specifications
Gaskets, Shellside	Per Project Specifications
Gaskets, Tubeside	Per Project Specifications
Gaskets, Nozzles	Per Project Specifications

ESTIMATED WEIGHTS

Weights per Shell (Tonnes)	Exchanger Weight, Dry	TBA	Exchanger Weight, Wet	TBA
	Bundle Weight, Dry	5.5		

DESIGN REQUIREMENTS

PED Classification	Shellside- N.A.	Tubeside- N.A.
Project Specifications	UOP 963766 - 401, UOP Std 4-11-7, UOP Std 3-11, UOP Std 3-15, UOP Std 9-11-7, 14780-21A1	
Design Code	ASME VIII Div 1	
Equipment Code	TEMA Class R 9th edition, API 660 8th edition 2007	



SHELL AND TUBE EXCHANGER THERMAL DATA SHEET

System of Units

Metric

Doc. No. : 14780-8120-PS-E2303

Rev. No. : Z1

Item No. : E-2303

Page : 5 / 10

Notes:

1. Item has been checked for Flow Induced Vibration. Tube damage is not predicted.
2. Flow and duty include 25% design margin.
3. Exchanger shall be installed horizontally, alongside E-2302A/B into the Preflash Column V-2301. Refer to UOP 963766-301 (Vessels). See page 6, 7.
Bases on detail "A" on page 7, CONTRACTOR to revise the drawing for a TEMA type C front head. OWNER/CONSULTANT approval is required if a TEMA A or B front head is used.
4. Additional requirements and design notes on page 7 shall be applied.
5. As per UOP 963766-401, the specified bundle outlet pressure is -0.9417 kg/cm²(g).
To meet the bundle outlet temperature of 70.1°C, the operating pressure at the bottom of bundle is -0.9210 kg/cm²g. VENDOR to check and confirm.
6. At normal flow, excluding 25% design margin, the calculated shellside pressure drop is 0.076 kg/cm².
7. Based on LMTD of 44.4°C for equal cold side temperature of 70.1°C at inlet and outlet, %over surface is 29.8%.
8. The effective finned area excludes U bend area and area of tubes at portion of bundle.
9. Alternate design steam conditions at inlet are 2.0 kg/cm²g at 142°C. Saturation temperature is 133°C.
10. Fire water may be used as an emergency coolant through the tube side. Design Pressure to be confirmed by CONTRACTOR.
11. The following information Based on low-fin tubes manufacturer's specification.

Fin density	1102.4	fin/meter
Plain average tube diameter	19.05	mm
Plain tube thickness (average)	1.651	mm
Fin root diameter	17.272	mm
Fin height	0.889	mm
Fin thickness	0.304	mm
Wall thickness under fins	1.067	mm

Low-fin tubes require no-land (Tube finned in baffles is required).

12. Strength-weld tubes to tubesheet per requirements as specified on page 8.
13. For U-tube design, Number of tubes equals number of straights.
14. Provide one in bend U-bend support to hold 9 outer most tube rows.
15. CONTRACTOR to confirm the required stabbed-in nozzle diameter of 1160 mm and provide vessel flange information/drawing by Exchanger VENDOR to Vessel VENDOR for flanges alignment.
16. Deleted.

Rev

SHELL AND TUBE EXCHANGER THERMAL DATA SHEET

System of Units Metric

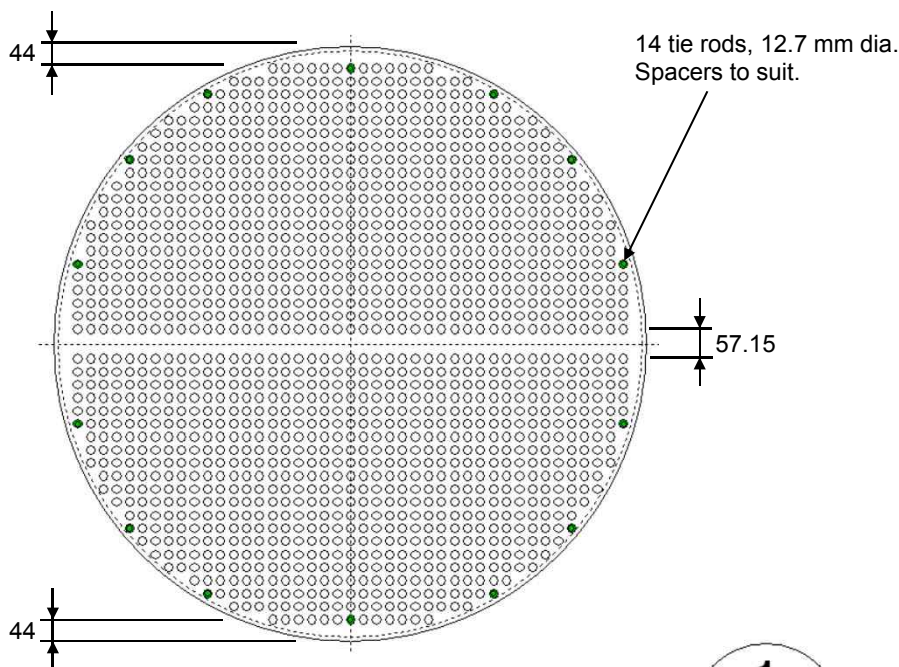
Doc. No. : 14780-8120-PS-E2303

Rev. No. : Z1

Item No. : E-2303

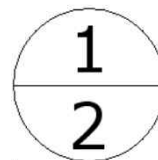
Page : 6 / 10

Tube Layout



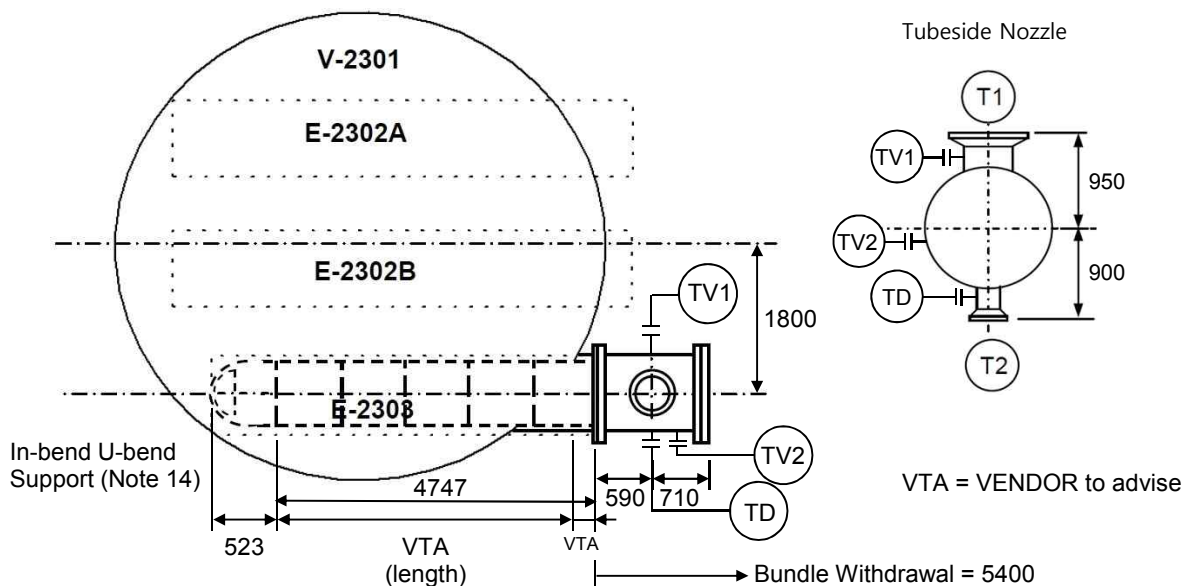
O.T.L. = 1142.7 mm

All dimension in mm
Sketch not to scale



Tubepass Arrangement

Setting Plan (Refer tp UOP 963766-301)



SHELL AND TUBE EXCHANGER THERMAL DATA SHEET

System of Units

Metric

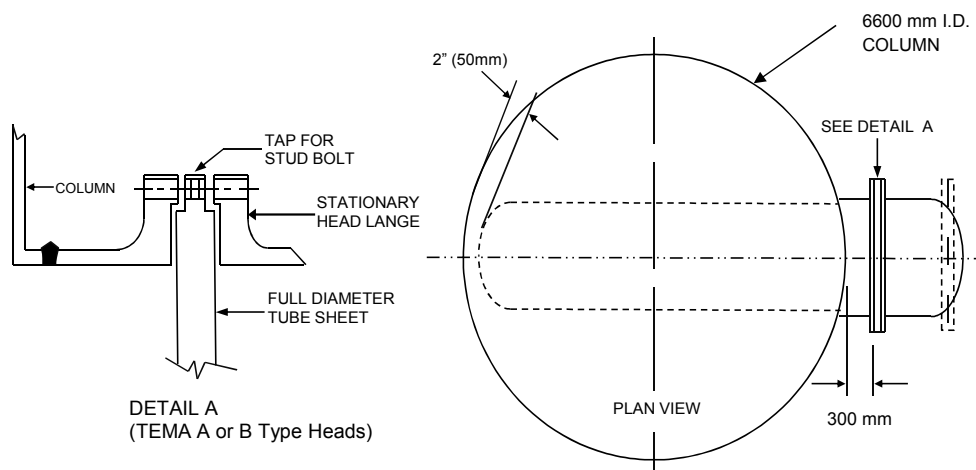
Doc. No. : 14780-8120-PS-E2303

Rev. No. : Z1

Item No. : E-2303

Page : 7 / 10

STABBED-IN REBOILER REQUIREMENTS (Note 4)



1. REQUIREMENTS

- Bundle vendor shall provide the necessary welding neck flange to column fabricator for attachment to column nozzle.
- Responsibility for design of detail "A" closure is fully chargeable to tube bundle designer except it shall be the TEMA "R" confined gasket type.
- Bundle vendor shall include stationary head together with necessary bolts, nuts and gaskets for securing bundle to column nozzle.

2. TUBE BUNDLE DESIGN:

- Tube bundle shall be U-tube type complying with above dimensions and with a nominal tube length that assures full traversing of column diameter per the dotted rectangle. U-tubes shall conform to UOP Standard Specification 4-11, Section 3.1. For HF Alkylation process applications where process stream contains acid, U-bends shall be stress relieved after bending per Section 4.4.a.12.
- Tube joints shall be the TEMA "R" double groove, rolled type. When welding of tube ends to tubesheet is required it will be noted on project item specification sheet.
- Occasionally this specification sheet will indicate a dimension from column wall to face of nozzle flange in excess of 12" (300 mm). This procedure is used for procuring an increase of bundle tube length at the expense of the neck extension versus the added cost of swaging the column size to a larger diameter. This, however, can be effective only when the tube bundle diameter is limited so there is an annulus in the connection neck that permits the liquid to flow in and the vapor to flow out of this portion of the bundle in an orderly manner.
- Bolt holes in tube sheet shall be drilled and tapped to form joint that permits removal of head while column connection to tube sheet joint remains tight. Bolt holes shall straddle vertical centerline.
- Column fabricator will provide two slide rail supports, for each bundle, installed inside column at 30° off bottom vertical centerline of bundle nozzle. Slide faces will be flush with surface of nozzle. Bundle support plate(s) shall be designed to slide on above rails.
- Provide a minimum of 1" (25 mm) clearance, above and below tube bundle, to permit flow into and out of portion of bundle that projects out from column.
- Design the tube bundle for vibration per UOP Standard Specification 4-11, Section 2.4.
- When a TEMA "C" type front head is specified, the bolting of the channel and integral tubesheet to the column nozzle flange shall be typical of a TEMA "C" type head to a shell.

Rev

SHELL AND TUBE EXCHANGER THERMAL DATA SHEET

System of Units Metric

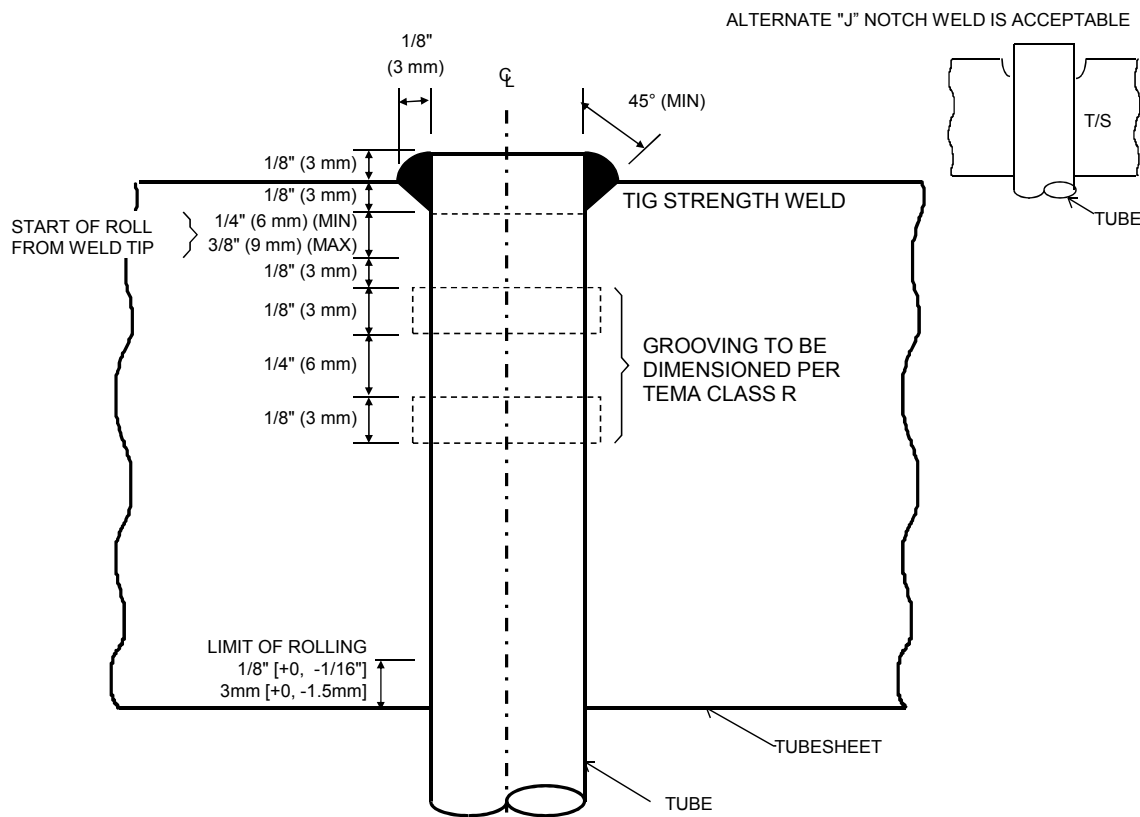
Doc. No. : 14780-8120-PS-E2303

Rev. No. : Z1

Item No. : E-2303

Page : 8 / 10

Strength-Weld Type Tube to Tubesheet Joint Requirements (Refer to UOP 963766 - 401)



1. Tubing shall be seamless type.
2. Joint components shall be deburred and meticulously cleaned prior to welding.
3. When heat treatment of the tubesheet is required, final tube hole sizing shall be attained by reaming after heat treatment.
4. Preheat tubesheet to a minimum of 150°F (65°C) and maintain this temperature during welding.
5. Each tube end shall be welded individually without stopping.
6. Tube to tubesheet joints shall be welded with a minimum of two passes. The first pass shall be 100% examined by dye penetrant or magnetic particle method. Remaining passes shall be welded without interruption. The final welds shall be 100% examined by dye penetrant or magnetic particle method.
7. All tube joint welds shall be proven using a soap test [with minimum 15 psi (1.05 kg/cm²) air] prior to rolling.
8. A helium leak test shall be performed in accordance with the requirements of ASME code, Section V, Article 10. The shell side shall be pressurized with helium while the exchanger is inclined to provide good flow of helium to a single collection point on the tube side. Any leakage in excess of 1×10^{-4} standard cc/sec warrants repair.
9. Do all tube rolling after completion of welding and leak testing.
10. The tubes shall be rolled for the entire tubesheet thickness. Stop shall be used on tube roller that assures expansion is within limits given on sketch.
11. A certified test report of the final, successful helium test shall be provided and signed by an independent third party witness.
12. All applicable Weld Procedure Specifications (WPS) and Welder Qualification Records (WQR) shall be provided per clients request.

Rev

SHELL AND TUBE EXCHANGER THERMAL DATA SHEET

System of Units Metric

Doc. No. : 14780-8120-PS-E2303

Rev. No. : Z1

Item No. : E-2303

Page : 9 / 10

ColdSide Property Curve 1

Pressure -0.9080 kg/cm²g

Temperature °C	Entire Stream Enthalpy kcal/kg	Entire Stream Mass Vapour fraction	Vapour Density kg/m ³	Vapour Viscosity cP	Vapour Therm. Cond. kcal/h.m.°C	Vapour Spec. Heat kcal/kg.°C
55.1	0.00	0.0000	-	-	-	-
68.8	5.47	0.0000	-	-	-	-
82.0	10.95	0.0000	-	-	-	-
94.8	16.42	0.0000	0.4741	0.008	0.0122	0.36
95.4	21.10	0.0536	0.4725	0.008	0.0123	0.36
96.2	25.79	0.1067	0.4725	0.008	0.0123	0.37
97.0	30.48	0.1593	0.4725	0.008	0.0124	0.37
98.0	35.16	0.2112	0.4709	0.008	0.0125	0.37
99.0	39.85	0.2624	0.4709	0.008	0.0126	0.37
100.2	44.54	0.3126	0.4693	0.008	0.0127	0.37

Temperature °C	Liquid Density kg/m ³	Liquid Viscosity cP	Liquid Therm. Cond. kcal/h.m.°C	Liquid Spec. Heat kcal/kg.°C	Liquid Surf. Tension dyne/cm	Critical Pressure kg/cm ² [a]	Critical Temperature °C
55.1	886.5	1.092	0.089	0.39	24.1	33.2	351
68.8	873.9	0.888	0.086	0.41	22.7	33.2	351
82.0	861.5	0.739	0.084	0.42	21.4	33.2	351
94.8	849.2	0.628	0.082	0.44	20.2	33.2	351
95.4	851.5	0.643	0.081	0.43	20.1	33.2	350
96.2	853.9	0.660	0.081	0.43	20.0	33.2	350
97.0	856.6	0.678	0.080	0.43	19.9	33.2	349
98.0	859.5	0.698	0.079	0.43	19.8	33.3	349
99.0	862.6	0.720	0.078	0.42	19.6	33.3	348
100.2	865.9	0.743	0.078	0.42	19.5	33.4	348

ColdSide Property Curve 2

Pressure -0.9556 kg/cm²g

Temperature °C	Entire Stream Enthalpy kcal/kg	Entire Stream Mass Vapour fraction	Vapour Density kg/m ³	Vapour Viscosity cP	Vapour Therm. Cond. kcal/h.m.°C	Vapour Spec. Heat kcal/kg.°C
55.1	0.00	0.0000	-	-	-	-
68.9	5.51	0.0000	-	-	-	-
82.2	11.01	0.0000	0.2995	0.007	0.0111	0.35
82.8	15.80	0.0542	0.2995	0.007	0.0112	0.35
83.6	20.59	0.1079	0.2979	0.007	0.0112	0.35
84.4	25.38	0.1611	0.2979	0.007	0.0113	0.35
85.4	30.17	0.2136	0.2979	0.007	0.0114	0.35
86.6	34.96	0.2654	0.2963	0.007	0.0115	0.36
87.9	39.75	0.3161	0.2963	0.008	0.0116	0.36
89.5	44.54	0.3657	0.2947	0.008	0.0118	0.36

Temperature °C	Liquid Density kg/m ³	Liquid Viscosity cP	Liquid Therm. Cond. kcal/h.m.°C	Liquid Spec. Heat kcal/kg.°C	Liquid Surf. Tension dyne/cm	Critical Pressure kg/cm ² [a]	Critical Temperature °C
55.1	886.5	1.092	0.089	0.39	24.1	33.2	351
68.9	873.8	0.887	0.086	0.41	22.7	33.2	351
82.2	861.3	0.738	0.084	0.42	21.4	33.2	351
82.8	863.8	0.758	0.084	0.42	21.3	33.2	350
83.6	866.5	0.782	0.083	0.42	21.2	33.2	350
84.4	869.4	0.808	0.082	0.42	21.1	33.3	349
85.4	872.6	0.837	0.081	0.41	21.0	33.3	349
86.6	876.2	0.869	0.080	0.41	20.8	33.3	348
87.9	879.9	0.904	0.079	0.41	20.7	33.4	347
89.5	883.9	0.942	0.078	0.41	20.5	33.4	346

Rev

SHELL AND TUBE EXCHANGER THERMAL DATA SHEET

System of Units Metric

Doc. No. : 14780-8120-PS-E2303

Rev. No. : Z1

Item No. : E-2303

Page : 10 / 10

ColdSide Property Curve 3

Pressure

-1.0031

kg/cm²g

Temperature °C	Entire Stream Enthalpy kcal/kg	Entire Stream Mass Vapour fraction	Vapour Density kg/m ³	Vapour Viscosity cP	Vapour Therm. Cond. kcal/h.m.°C	Vapour Spec. Heat kcal/kg.°C
55.1	0.00	0.0000	-	-	-	-
57.2	0.80	0.0000	-	-	-	-
59.2	1.60	0.0000	0.1169	0.007	0.0091	0.33
60.0	7.74	0.0681	0.1169	0.007	0.0092	0.33
60.9	13.87	0.1356	0.1153	0.007	0.0093	0.33
62.0	20.00	0.2022	0.1153	0.007	0.0094	0.33
63.4	26.14	0.2678	0.1153	0.007	0.0095	0.33
65.1	32.27	0.3318	0.1153	0.007	0.0096	0.34
67.4	38.41	0.3937	0.1137	0.007	0.0099	0.34
70.3	44.54	0.4528	0.1137	0.007	0.0101	0.34

Temperature °C	Liquid Density kg/m ³	Liquid Viscosity cP	Liquid Therm. Cond. kcal/h.m.°C	Liquid Spec. Heat kcal/kg.°C	Liquid Surf. Tension dyne/cm	Critical Pressure kg/cm ² [a]	Critical Temperature °C
55.1	886.5	1.092	0.089	0.39	24.1	33.2	351
57.2	884.6	1.058	0.088	0.40	23.9	33.2	351
59.2	882.8	1.025	0.088	0.40	23.7	33.2	351
60.0	886.2	1.072	0.087	0.40	23.6	33.2	350
60.9	890.2	1.128	0.086	0.39	23.5	33.2	349
62.0	894.7	1.195	0.085	0.39	23.3	33.3	349
63.4	899.8	1.274	0.084	0.39	23.1	33.3	348
65.1	905.6	1.366	0.083	0.38	22.9	33.4	347
67.4	912.1	1.472	0.081	0.38	22.6	33.5	346
70.3	918.9	1.585	0.079	0.37	22.2	33.5	344

Rev